Stabilizing Training by Capping Local Representation Displacement

An intervention motivated by destructive-regime tracing

Analysis 065 / H048 8 November 2026

Findings at a glance

We translate layerwise representation-instability tracing into an intervention: shrink a neuron's proposed Adam increment only when its direct affine response on a fixed probe exceeds a past-calibrated budget. All layers remain trainable. Distributed caps reduce mean training loss by **22.85%, 16.22%, and 26.00%** for MLP/CIFAR-10, ResNet/CIFAR-10, and Transformer/WikiText-2 in the selected windows. Two further MLP initializations improve by 19.43% and 32.08%. Loss excursions shrink even when representation reversals persist or increase. **Terminal test loss worsens under the distributed cap in every setup.** These are finite-window optimization results, not a generalization or convergence guarantee.

1. Motivation: from destructive diagnostics to intervention

At state t (after t completed updates), let A_t^ℓ be a layer's representation on a fixed ordered probe. Rows retain examples or token/spatial positions; columns retain features. For P_ℓ scalar entries, define

$$\delta_t^\ell = \mathrm{vec}(A_{t+1}^\ell - A_t^\ell), \qquad \rho_t^\ell = \frac{\langle \delta_t^\ell, \delta_{t-1}^\ell \rangle}{\|\delta_t^\ell\|_2 \, \|\delta_{t-1}^\ell\|_2}, \qquad r_t^\ell = \frac{\|\delta_t^\ell\|_2}{\sqrt{P_\ell}}. \tag{1}$$

The raw full-objective gradient is g_t, with $\rho_t^g = \cos(g_t, g_{t-1})$. We also report $\eta\widehat{\kappa}_t = 1 - \langle g_t, g_{t-1}\rangle / \|g_{t-1}\|_2^2$: normalized projected gradient change, not an Adam Hessian eigenvalue or stability threshold. A zero denominator makes a cosine or this ratio undefined. Negative cosine describes adjacent cancellation, termed *destructive* geometry; it does not imply that loss must rise. Earlier backward tracing found large reversing changes already at the first MLP projection, distributed through ResNet stages, and after attention despite steadier Transformer embeddings. The temporal SVD of stacked δ_t^ℓ revealed concentrated motion. Its window-fitted basis is fixed by construction; instantaneous dominant directions can still rotate. This descriptive evidence motivates controlling movement where it is produced, without identifying a unique initiating layer.

Our operational stabilization target is smaller loss excursions while preserving training progress. We measure the maximum and total positive loss variation alongside mean and tail loss. The controller acts on displacement amplitude; it neither triggers on a cosine threshold nor uses future-window SVD vectors. Thus it is *informed by* destructive-regime diagnostics, rather than treating their sign as an objective or a stability theorem.

2. Local displacement: the measured quantity

For one affine module, let $H_t \in \mathbb{R}^{n\times d}$ contain its current probe inputs, $W_t \in \mathbb{R}^{m\times d}$ its weights, and $b_t \in \mathbb{R}^m$ its biases. Its preactivation matrix is $Z_t = H_t W_t^\top + \mathbf{1} b_t^\top$. Adam proposes stored parameter increments U_t and v_t of matching shapes. Holding H_t fixed, compute

$$D_t = H_t U_t^\top + \mathbf{1} v_t^\top \in \mathbb{R}^{n\times m}, \qquad s_{t,j} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} D_{t,ij}^2}. \tag{2}$$

Column j of D_t contains the direct changes of neuron j across all probe rows. For MLP linear 1, H_t is 5000×3072 and D_t is 5000×200. The score is RMS down a column: examples are squared before averaging, so their signs cannot cancel. It is not a worst-case bound on each example.

3. Update rule and experimental design

For each targeted module ℓ, pool $s^{\ell}_{t,j}$ over its units and the eight untreated updates preceding the fork t_0. Set τ_ℓ to their chosen quantile q, and keep it fixed. Current inputs and scores are recomputed each step. For a positive threshold,

$$a^{\ell}_{t,j} = \min(1, \tau_\ell / s^{\ell}_{t,j}), \qquad \widetilde{U}^{\ell}_{t,j:} = a^{\ell}_{t,j} U^{\ell}_{t,j:}, \qquad \widetilde{v}^{\ell}_{t,j} = a^{\ell}_{t,j} v^{\ell}_{t,j}. \tag{3}$$

Use $a = 1$ for a zero score. A zero calibrated threshold disables that module's cap. The same multiplier scales a neuron's full weight row and bias; all untargeted increments remain native. Its direct RMS becomes $a_{t,j} s_{t,j} \le \tau$ in exact arithmetic; storage rounding is checked separately. q50 denotes a calibration median, not a requirement to modify half the units at each subsequent step.

Algorithm 1 Local representation displacement cap at update t

Require: Parameters θ_t, Adam state, fixed probe, target modules $\mathcal{S}$, budgets τ_ℓ

1: Capture current module inputs H^{ℓ}_t on the probe; retain θ_t.
2: Compute the raw gradient g_t of the complete training objective.
3: Advance native Adam moments/clocks and form stored proposal θ^{+}_t.
4: **for** each targeted module $\ell \in \mathcal{S}$ with $\tau_\ell > 0$ **do**
5: Extract proposed increments (U^{ℓ}_t, v^{ℓ}_t) from $\theta^{+}_t - \theta_t$.
6: Compute direct affine responses and per-unit RMS using Eq. (2).
7: Scale each proposed weight row and bias using Eq. (3).
8: **end for**
9: Store $\theta_{t+1} = \theta_t + \widetilde{u}_t$; keep the advanced Adam state.
10: Measure loss and actual consecutive representation changes for evaluation.

Entry versus distributed. The entry variant targets MLP linear 1, ResNet conv 1, or the first Transformer QKV projection. The distributed variant targets both MLP hidden affines, all 31 ResNet convolutions, or both QKV projections plus all four feed-forward affines. MLP output weights, ResNet normalization/classifier parameters, and Transformer attention output projections, LayerNorm, embeddings and decoder are outside the cap. A convolutional score averages over images and spatial positions; QKV scores measure the projection, not the complete attention operation. Selection is over existing units/channels, not temporal singular vectors.

Case	Full objective per update	Fixed probe	LR; fork; final state
MLP / CIFAR-10	5,000 images; half squared output error	All 5,000 images, native 1,000-image chunks	.001; 448; 800/1152
ResNet / CIFAR-10	50,000 images; mean cross-entropy	First intact 1,000-image BatchNorm group	.003; 264; 576
Transformer / WikiText-2	2,088,600 valid tokens; mean NLL	First 64 length-35 sequences: 2,240 valid positions	.003; 552; 928

Table 1: Full-objective deterministic updates throughout. Probe coverage changes only measurement and the local constraint, not the training objective.

The MLP is $3072 \to 200 \to 200 \to 10$ with two tanh activations and a linear output. ResNet-32 has stage widths 16/32/64 and BatchNorm without running statistics. The two-block Transformer has width 200, two attention heads, feed-forward width 200, zero dropout and untied embeddings/decoder (vocabulary 33,278). All use Adam $(\beta_1, \beta_2) = (.9, .999)$, $\epsilon = 10^{-8}$, no weight decay or clipping, FP32 parameters and disabled TF32. Dataset order, cached data, initialization and source/config hashes are pinned. All gradients and moments continue to update; future gradients naturally depend on the intervened trajectory. For the MLP, $L = (2N)^{-1} \|F - Y\|^2_F$ with one-hot targets Y. Output diagnostics retain raw linear outputs for MLP and all class/vocabulary log-probabilities for ResNet/Transformer, without averaging probe rows.

4. Results: cap variants and unchanged-Adam controls

The primary entry screen uses q75; the common distributed comparison uses q50. Each branch is paired with unchanged Adam from its exact native fork on the same worker. Mean loss averages the stated loss states; tail averages the last 32 states; PV $= \sum_t \max(L_{t+1} - L_t, 0)$ uses adjacent states within that window. Outgoing motion also uses a retained state after its end. Maximum includes the shared initial state. Percent changes below are paired descriptive effects; no confidence intervals are claimed.

Model	Cap	States	Mean	Tail	Maximum	PV
MLP	Entry q75	448–800	−8.26%	+8.50%	−37.91%	−69.50%
ResNet	Entry q75	264–576	−6.29%	−22.76%	+21.44%	−17.15%
Transformer	Entry q75	552–928	−8.53%	−24.58%	−14.68%	−45.20%
MLP	Distributed q50	448–1152	−22.85%	+4.36%	−39.04%	−85.49%
ResNet	Distributed q50	264–576	−16.22%	−7.48%	−13.82%	−46.78%
Transformer	Distributed q50	552–928	−26.00%	−73.36%	−79.63%	−98.81%

Table 2: Percent changes relative to each branch's own unchanged Adam control. Negative values mean smaller loss or positive variation. MLP distributed results use the extended window; the entry screen stops at 800.

Entry caps improve the mean, but not every outcome. The q75 entry rule lowers average loss on all three setups, yet increases the MLP tail and ResNet maximum. It cannot be called a uniform suppression of loss peaks. **Distributed q50 improves mean loss and PV in every setup.** Its MLP tail is 4.36% worse, while ResNet and Transformer tails improve. Figure 1 shows entry caps; Figure 2 shows distributed caps, including raw-gradient and actual representation diagnostics.

Model	Additional cap variant	Mean	Tail	PV
MLP	Entry q50	−20.94%	+13.56%	−82.03%
MLP	Distributed q75	−8.44%	+35.32%	−71.44%
ResNet	Distributed q75	−17.42%	−8.31%	−17.71%

Table 3: Scope/quantile checks using the full respective windows (MLP 448–1152; ResNet 264–576). MLP entry q50 and distributed q50 share a matched worker; ResNet entry q75 and distributed q75 are also matched.

The matched-q50 MLP comparison favors distribution in mean loss (−22.85% versus −20.94%); the matched-q75 ResNet comparison does too (−17.42% versus −6.29%). These are joint changes in target scope and the resulting applied updates, not equal-norm or equal-cost comparisons. The Transformer entry and distributed experiments also differ in quantile and GPU architecture (RTX 4090 versus RTX 5090). Their control trajectories differ; we compare each cap only with its own control and do not claim an isolated Transformer depth effect. The setup comparisons are not an architecture-only ablation.

Stabilization can retain destructive geometry. Under distributed q50, the fraction of negative output-change cosines rises from 36.1% to 66.5% on ResNet and from 36.6% to 97.9% on Transformer. ResNet's mean output RMS rises from 0.567 to 0.723 while its maximum falls from 2.785 to 1.592. Transformer maximum output RMS falls from 15.365 to 1.834. Thus reduced excursions do not require forward-only motion or smaller movement at every step.

Held-out loss limits the claim. At the terminal state, distributed q50 raises test loss by 2.87% (MLP seed 0), 8.18% (ResNet), and 100.17% (Transformer). CIFAR-10 evaluation covers all 10,000 test images; WikiText-2 covers all 245,550 supervised test tokens, with token weighting. These are endpoint comparisons and may reflect different oscillation phases; no held-out trajectory or generalization improvement is established. MLP replication is reported on the last page.

Evidence and reproduction. Tracing motivation: Analysis 063 technical report. Cap evidence: H048 Runs 269–274 and 282–283, with exact fork/native continuation qualification, hash-verified retrieval and retained-state audits. This report uses only existing verified cap/control records; no new training. Within Analysis 065, run `34_local_cap_report_assets.py`, then `35_build_local_cap_report.py`. Source hashes, table reductions, plotted values and PDF review are recorded in the accompanying report verification files.

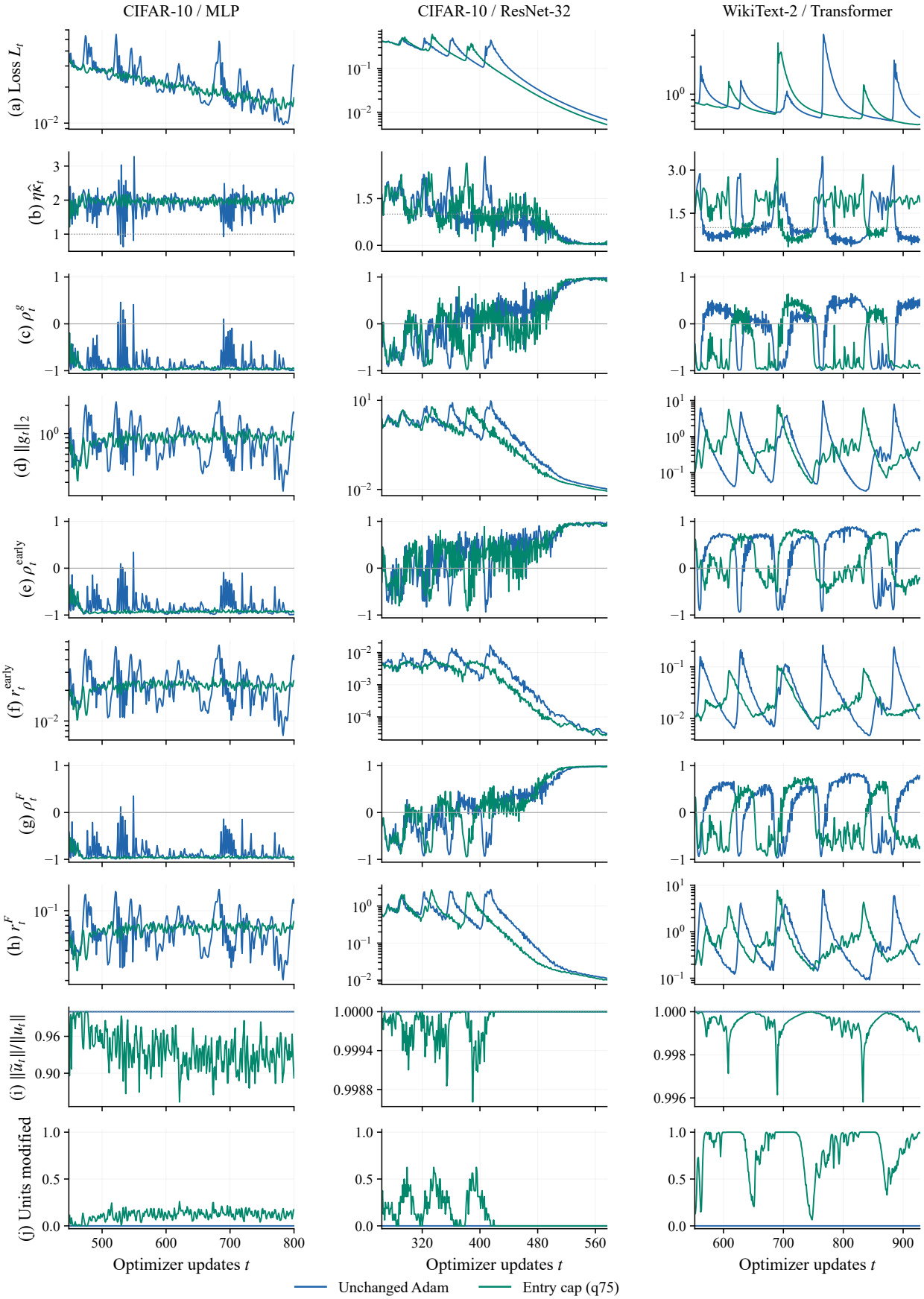


Figure 1: Entry local displacement caps (q75). Each column compares only the cap with unchanged Adam on its own worker. Loss, gradient norm and displacement RMS use logarithmic axes; $\eta\widehat{\kappa}_t$ uses a linear axis. Magnitude scales are independent by column. Early representation means first MLP tanh, ResNet conv 1 output, or first Transformer attention output. ρ^g uses consecutive raw gradients; ρ^{early} and ρ^F use consecutive outgoing representation changes. Row (i) divides the actual whole-parameter increment norm by the native proposal norm. Row (j) averages fractions of modified units across targeted modules. All curves are unsmoothed; zero-denominator cosines are gaps.

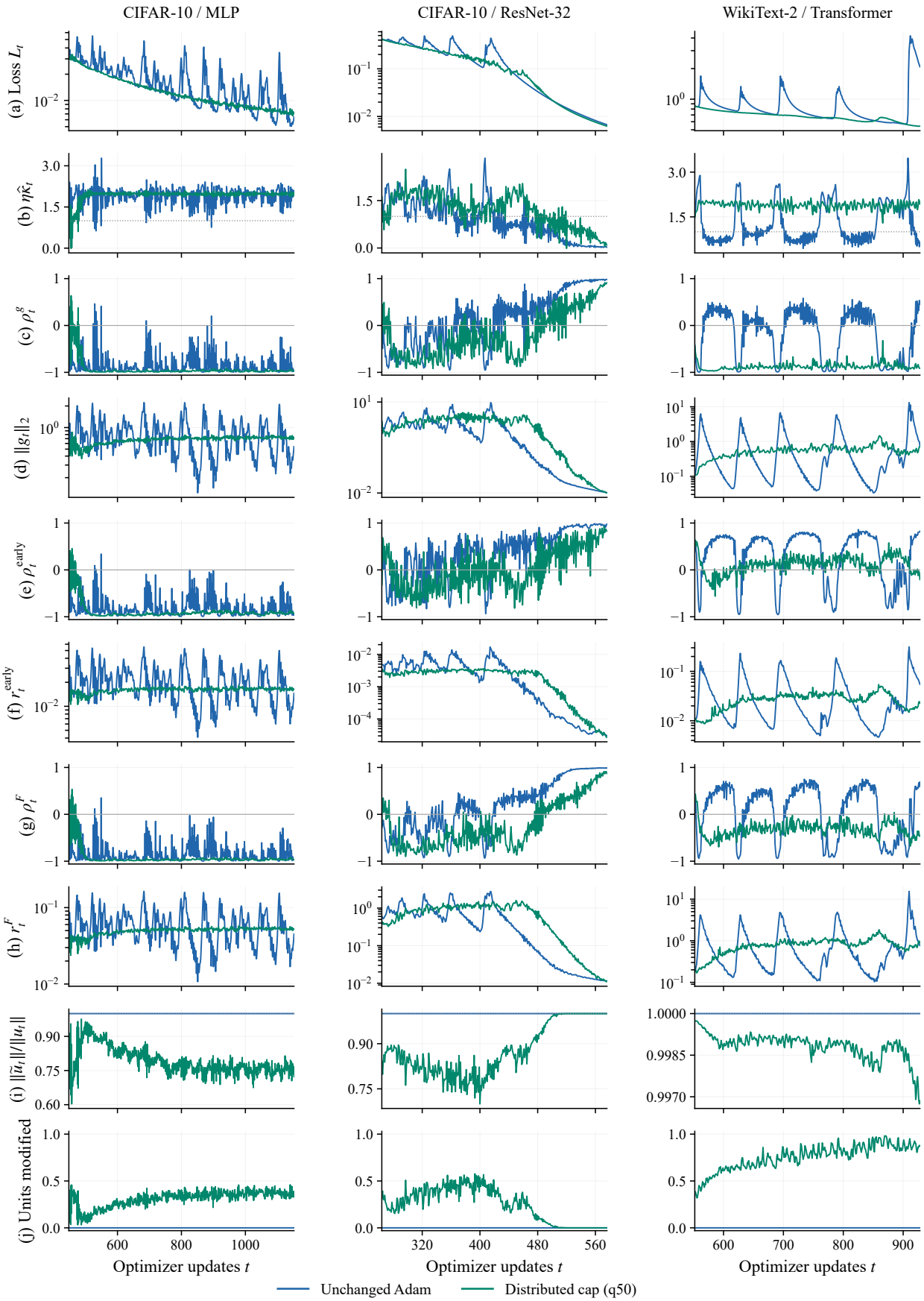


Figure 2: Distributed local displacement caps (q50). Same diagnostics and independent column scales as Figure 1; MLP is extended through 1152. Loss excursions and mean loss decline while output reversals persist. The Transformer cap targets only 402,000 of 13,828,478 parameters (2.91%), so the whole-update norm ratio stays close to one; it is not the fraction of targeted units modified. This controller bounds direct local contributions, not complete-network output displacement. Transformer baselines in this figure and Figure 1 come from different GPU architectures and must not be treated as the same numerical trajectory.

5. Replication, interpretation and practical limits

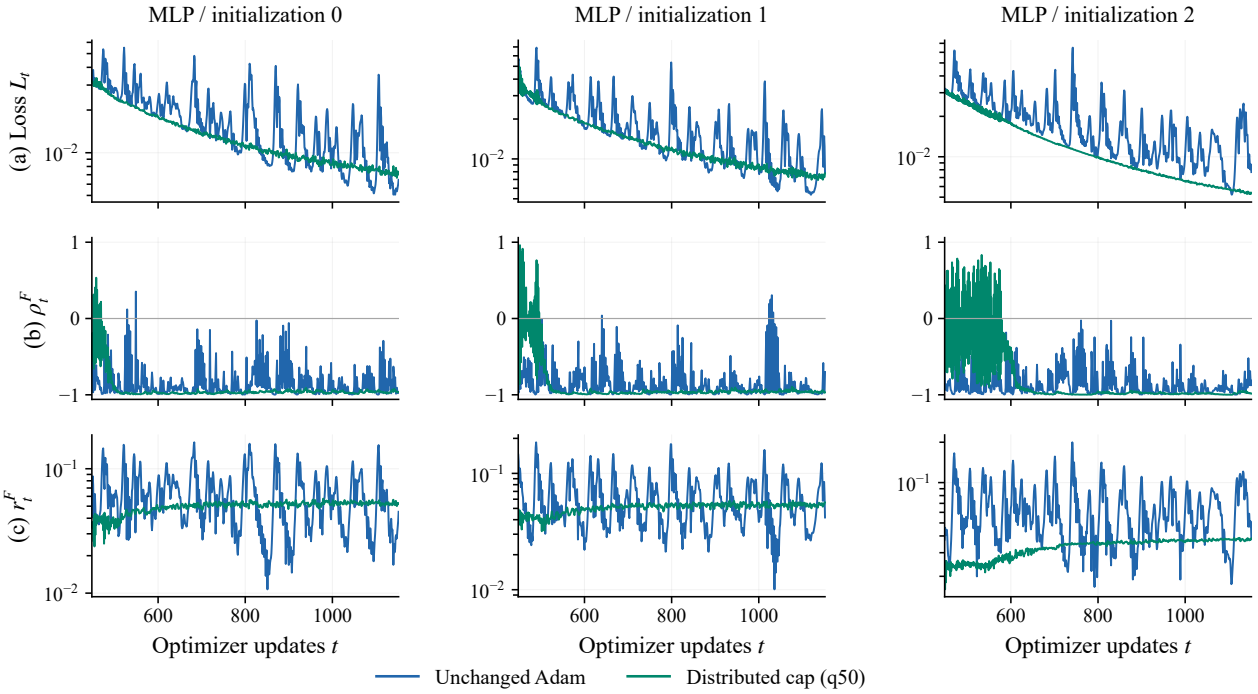


Figure 3: MLP distributed-q50 replication, states 448–1152. Initialization 0 selected the candidate; initializations 1/2 received the same rule without retuning. Loss and output RMS use logarithmic axes with independent scales per column. Each initialization has its own unchanged Adam control.

Initialization	Mean training loss	Tail loss	PV	Terminal test loss
0	−22.85%	+4.36%	−85.49%	+2.87%
1	−19.43%	−28.52%	−73.92%	+2.76%
2	−32.08%	−64.49%	−82.56%	+6.13%

Table 4: Percent changes under distributed q50. Better average training loss replicates; better terminal test loss does not.

What is bounded. Because upstream layers also move, the actual affine change is

$$\Delta Z_t = \underbrace{H_t \widetilde{U}_t^\top + \mathbf{1}\widetilde{v}_t^\top}_{\text{capped direct contribution}} + \underbrace{\Delta H_t W_t^\top + \Delta H_t \widetilde{U}_t^\top}_{\text{incoming movement and interaction}} . \tag{4}$$

Only the first term is constrained locally. Nonlinearities, normalization and attention further transform it. For the first MLP layer, the input is fixed, so $\Delta H_t = 0$. These caps preserve each unit's direct displacement direction while changing its magnitude; differing factors can change the combined direction.

Cost and scope. The implementation captures probe inputs, evaluates direct affine products in FP64 and checks final FP32 rounding. It requires no Jacobian or SVD. Instrumented control/cap seconds per update are 0.01376/0.05797 (MLP), 3.53170/3.83282 (ResNet), and 7.80567/7.95608 (Transformer). Concurrent Spark work makes the MLP timings non-isolated; these include diagnostics, audit writes and terminal evaluation, not production optimizer overhead. Fixed prefix probes and preservation of ResNet's native normalization group are essential; future padded probes need valid-position masking in local scores too.

Conclusion. Selective representation-amplitude control is a promising stabilization intervention in these deterministic full-objective continuations. Three MLP initializations and one per larger model support a limited transfer result. Selected forks, local calibration, unequal windows and the absence of stochastic-minibatch tests limit generality. The next validation should compare additional forks/seeds and repeated held-out measurements at matched training progress.